

Does harbor activity influence *Tridacna maxima* giant clam size, density and population dynamics on Heron Island?

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Abstract

Giant clams (*Tridacna maxima*) are important contributors to coral reef ecosystems, however harbor construction and boat activity are suggested to alter their ecological role by changing water flow, nutrient availability and sedimentation rates. To determine how harbor activity affects the *T. maxima* giant clam on Heron Island, 50m transect sampling was conducted across a disturbed site (the Harbor) and two undisturbed sites (Shark Bay and Research Beach) to compare clam density, shell length, weight, dry tissue biomass, and age composition. While clam density had no significant differences across the disturbed and undisturbed sites ($p = 0.51$), *T. maxima* had larger shell lengths, weights and dry tissue biomass at the Harbor than at either undisturbed site ($p < 0.05$). Additionally, the Harbor had a larger proportion of adults than juveniles compared to the two undisturbed sites ($p = 0.001874$). These patterns are likely due to increased nutrient availability along the Harbor through increased water flow, which favors growth but reduces juvenile recruitment. Larger juvenile populations in undisturbed sites suggest an increase in growth filtration rates and reef contribution through bioerosion. Additional research with targeted measurements of nutrient availability, calcification and filtration rates is needed to clarify *T. maxima* contributions to coral reef ecosystems on Heron Island.

Introduction

Bivalves are important reef-building organisms in supporting coral reef productivity and health. Among these are giant clams, which play vital ecological roles in coral reef productivity, diversity, and structural integrity (AbouElmaaty *et al.* 2024). Giant clams provide home to marine organisms like fish and invertebrates as their shells contribute to reef structure and topography (Neo *et al.* 2015). As filter feeders, giant clams sequester excess nutrients through biofiltration, and in turn help maintain the required oligotrophic nature of the reef ecosystem (Triandiza *et al.* 2019).

Giant clams within the Tridacninae subfamily function as photoautotrophs (Mohammed *et al.* 2019), contributing to reef primary productivity through symbiotic relationships with zooxanthellae (Neo *et al.* 2015). The zooxanthellae live within the clam's mantle (Griffiths and Klumpp 1996) and perform photosynthesis to provide the clam with the nutrients and energy needed for calcification (the formation their calcium carbonate shells) (Griffiths and Klumpp 1996; Neo *et al.* 2015). Increase in mantle size and shell length increases the number of zooxanthellae, however, symbiotic algal density - number of algae per gram of tissue - decreases as clams grow larger (Griffiths and Klumpp 1996). Light then must penetrate more tissue to reach the zooxanthellae, leading to lower growth rates among larger clams (Rossbach *et al.* 2019). However, because giant clams have larger mantles on average than other bivalves, they are among the fastest growing with high calcification rates (Rossbach *et al.* 2019).

Tridacna maxima is one of the smallest giant clam species within the subfamily Tridacninae (AbouElmaaty *et al.* 2024), however it has one of the largest rates of primary productivity compared to other reef organisms (Neo *et al.* 2015). These clams rely on light for growth (Rossbach *et al.* 2019), which results in a preference for shallow, clear-water intertidal habitats (Lee *et al.* 2024; Hart *et al.* 1998). These clams live embedded into hard substrate,

including but not limited to rocks, dome-shaped coral, or coral rubble (Gilbert *et al.* 2006; Triandiza *et al.* 2019; Andréfouët *et al.* 2009). *T. maxima* larvae settle on these structures by burrowing into the substrate (Rippe *et al.* 2024), and in turn contribute to the bioerosion and sedimentation in reefs (Neo *et al.* 2015). *T. maxima* has one of the fastest recruitment rates out of all giant clam species (Neo *et al.* 2015), thus their effects on reef contribution are worth studying.

Varying environmental conditions, including light, substrate availability, and water flow, impact giant clam growth, juvenile recruitment, and survival, like *T. maxima*. Human activity such as port construction can alter reef hydrodynamics and sedimentation, and in turn affect *T. maxima* populations (Van Wynsberge *et al.* 2016). Increased sedimentation and changes in tidal flow associated with harbor development reduce proper substrate availability for juvenile settlement and recruitment (Rippe *et al.* 2024; Triandiza *et al.* 2019). Heron Island has a single dredged port that allows boat access, which has been associated with increased tidal outflow, wave action, and sediment deposition on the surrounding reefs (Gourlay 1995). Changes in environmental conditions from the island's harbor provide disturbance to local reef ecosystems; however, their effects on *T. maxima* populations on Heron Island are unknown.

Previous studies indicate various impacts increased sedimentation and water movement have on *T. maxima* populations. In addition to lower juvenile recruitment (Rippe *et al.* 2024; Triandiza *et al.* 2019), Hart *et al.* 1998 suggests a decrease in growth rate among *T. maxima* in conditions with low visibility caused by sediment runoff and increased turbidity from tidal outflow. However, increased sediment runoff suggests nutrient enrichment, which may enhance shell growth and calcification rates (Ambariyanto 2007; Mills *et al.* 2025; Rippe *et al.* 2024). In contrast, Gilkinson *et al.* 2005 found a reduction in biomass across bivalves in areas of high dredging and sediment deposition (Gilkinson *et al.* 2005), and thus a decrease in carbonate

productivity (Neo *et al.* 2015). Conflicting suggestions for the harbor's effect on local *T. maxima* giant clams highlights the need for further research.

This study aims to answer the question if the disturbance caused by the Harbor affects the size, biomass, density, and population dynamics of *T. maxima* on Heron Island. Specifically, it provides an exploratory analysis of the differences of each variable across a disturbed site (the Harbor), and two undisturbed sites (Shark Bay and Research Beach) to determine the Harbors impact on *T. maxima* populations. This study also suggests the implications of the clam's contribution to coral reef ecosystems around Heron Island.

Methods

This study was conducted between October 24-26, 2025, across three sites on Heron Island, Great Barrier Reef. Sites included the Harbor (disturbed site) and Research Beach and Shark Bay (undisturbed sites). The Harbor represents the disturbed site due to the dredging and continuous boat activity, while the other two sites are defined as undisturbed due to minimal infrastructure and low vessel activity.

Sampling was conducted at low tide using four 50-meter transects at each site, following the methodology of Triandiza *et al.* 2019. Transects were non-random and began where the first *T. maxima* clam was observed, and in the direction of more clams to ensure a large sample size. All *T. maxima* within 5 meters to the left and right of the transect line were counted and identified morphologically using guides (Calumpong 1992; Mohammed *et al.* 2019). Each individual's picture and GPS coordinates were taken using GoPros, along with the meter mark of the transect to prevent the counting of the same clam twice. Due to a technological malfunction,

not all coordinates were recorded; however, the estimated distribution of clams surveyed across each site is shown in Fig. 1.



Fig. 1. Recorded coordinates of the giant clam *T. maxima* measured across three days of data collection in the Harbor, Research Beach, and Shark Bay.

Shell length was measured in situ from anterior to posterior edges using calipers (Armstrong *et al.* 2022; Pearson and Munro 1991) (Fig. 2). Respective shell weight and dry tissue biomass were estimated by calculation based on shell length measurements. Shell weight (g) was determined using the equation: $10^{[(3.1335 \times \text{LOG}(\text{SL}_{\text{cm}})) - 0.9173]}$ (Gilbert *et al.* 2006). Dry tissue biomass was also calculated, by taking 5.8% of the wet tissue biomass (Ricciardi and Bourget 1998), derived from the equation: $10^{[(3.0434 \times \text{LOG}(\text{SL}_{\text{cm}})) - 1.6026]}$

(Gilbert *et al.* 2006). Using shell length, clams were classified as juveniles (< 15 cm shell length) and adults (>15 cm shell length) as determined by Lee *et al.* 2024. Percentage of adults to juveniles were also recorded. Additionally, average density (individuals per 100 m²) across all four 50x10m transects at each respective site was calculated.

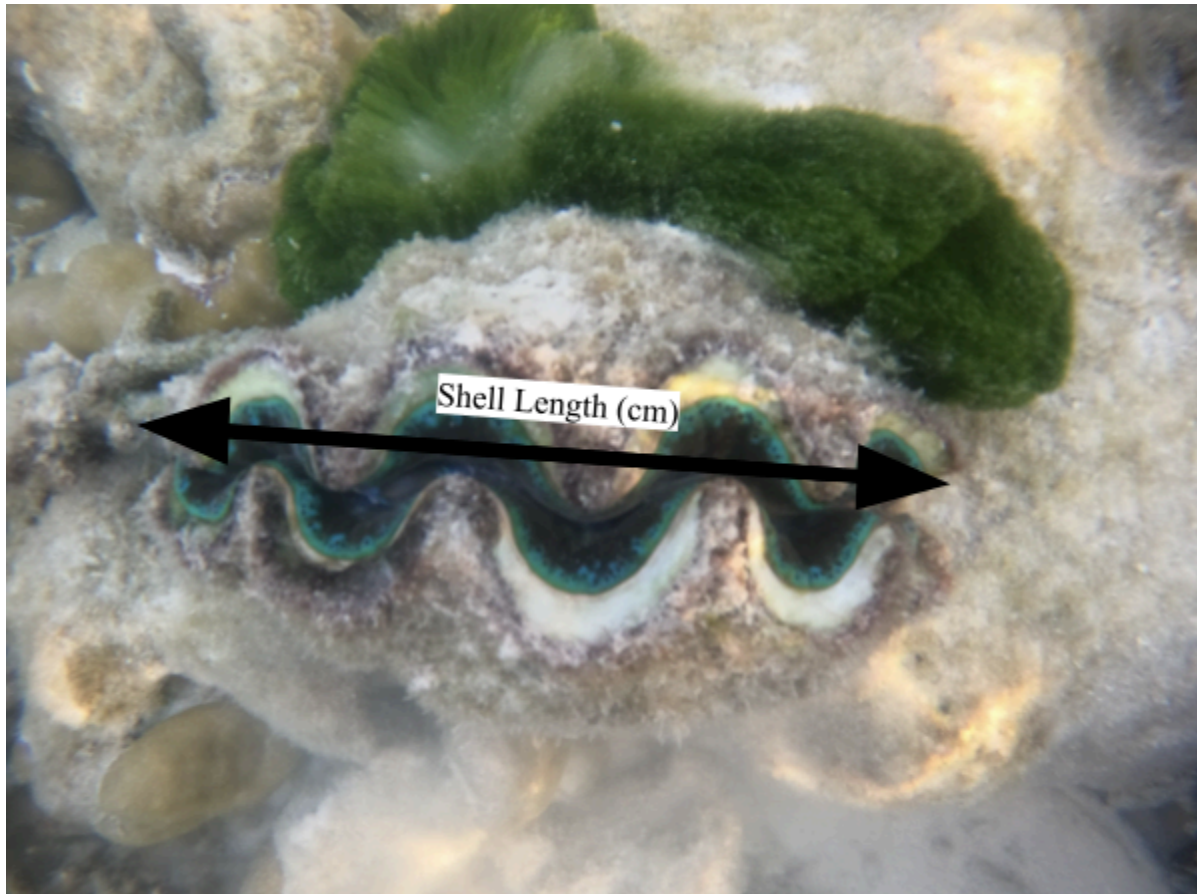


Fig. 2. Morphological shell length measurement of the in-situ *T. maxima* clam shell from anterior to posterior ends taken using calipers.

Statistical analysis was completed using RStudio (version 4.5.1). Differences in average density across sites were tested for significance using a Kruskal-Wallis test. Data were tested for normality and homogeneity of variance using Shapiro-Wilk and Levene's tests, respectively. If assumptions of normality and variance were met a one-way ANOVA and a Post-Hoc Tukey test

was used to compare shell length, weight, and biomass among sites. If assumptions were violated, a Kruskal-Wallis and a Pairwise Wilcox test were used. Differences in juvenile-to-adult ratios were tested using Pearson's Chi-square test.

Results

Average density ($\pm$ SD) of *T. maxima* per transect was 2.45 ± 2.163 individuals/100 m² at the Harbor, 1.65 ± 0.3 individuals/100 m² at Research Beach and 1.65 ± 0.473 individuals/100 m² at Shark Bay (Fig. 3). However differences among sites were not statistically significant (Kruskal-Wallis $\chi^2(2) = 1.33$, $p = 0.51$).

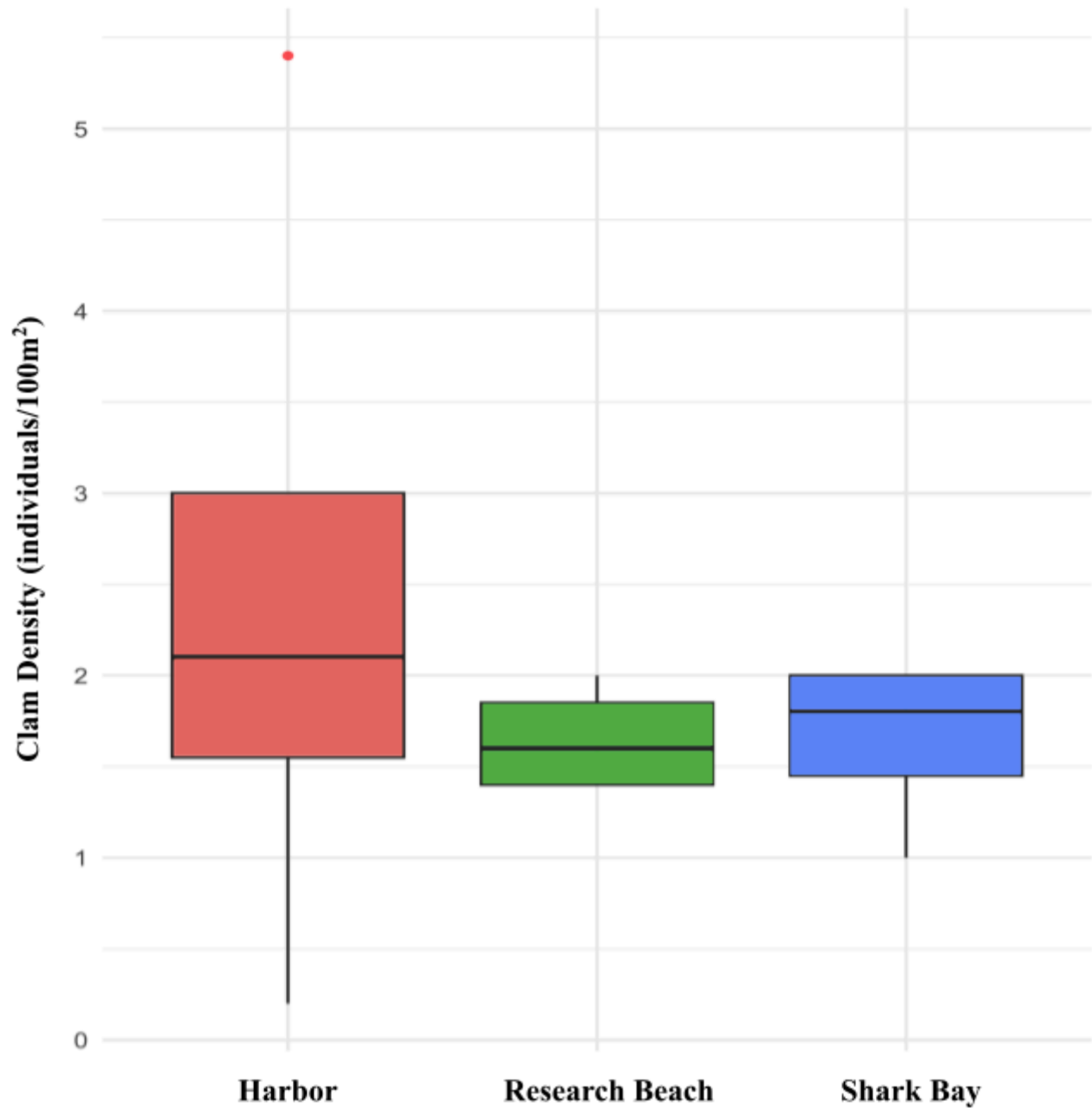


Fig. 3. Comparison of calculated mean density per transect (individuals/100 m²) of *T. maxima* across the three sites: the Harbor (disturbed), Research Beach and Shark Bay (undisturbed).

Mean shell length differed significantly across the three sites (ANOVA, $F(2, 112) = 8.55$, $p = 0.0004$). *T. maxima* shell length at the Harbor were significantly larger than those at Research

Beach ($p = 0.001$) and Shark Bay ($p = 0.005$), as determined by a post hoc Tukey test. No real difference was found between Research Beach and Shark Bay ($p = 0.889$) (Fig. 4).

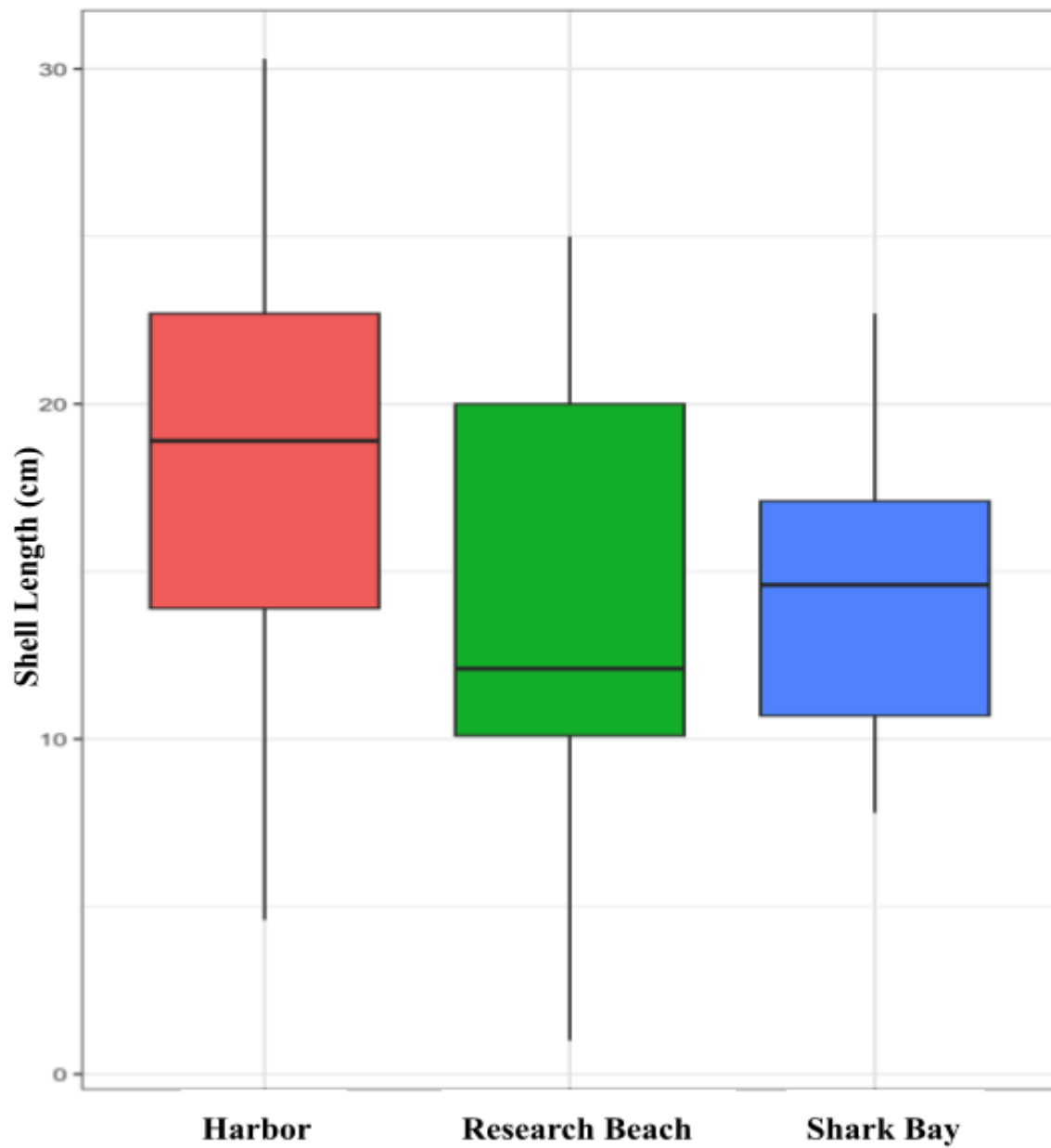


Fig. 4. Comparison of measured shell length of *T. maxima* across the three sites: the Harbor (disturbed), Research Beach and Shark Bay (undisturbed).

Shell weight data violated assumptions of normality and variance, thus Kruskal-Wallis test was used to show significant difference in weight across all three sites (Kruskal-Wallis, $\chi^2(2) = 14.27$, $p = 0.0008$). Approximate weight was significantly higher at the Harbor compared to Research Beach ($p = 0.0027$) and Shark Bay ($p = 0.0027$), as determined by a Pairwise Wilcoxon test. Shell weight did not significantly differ between Research Beach and Shark Bay ($p = 0.49$) (Fig. 5).

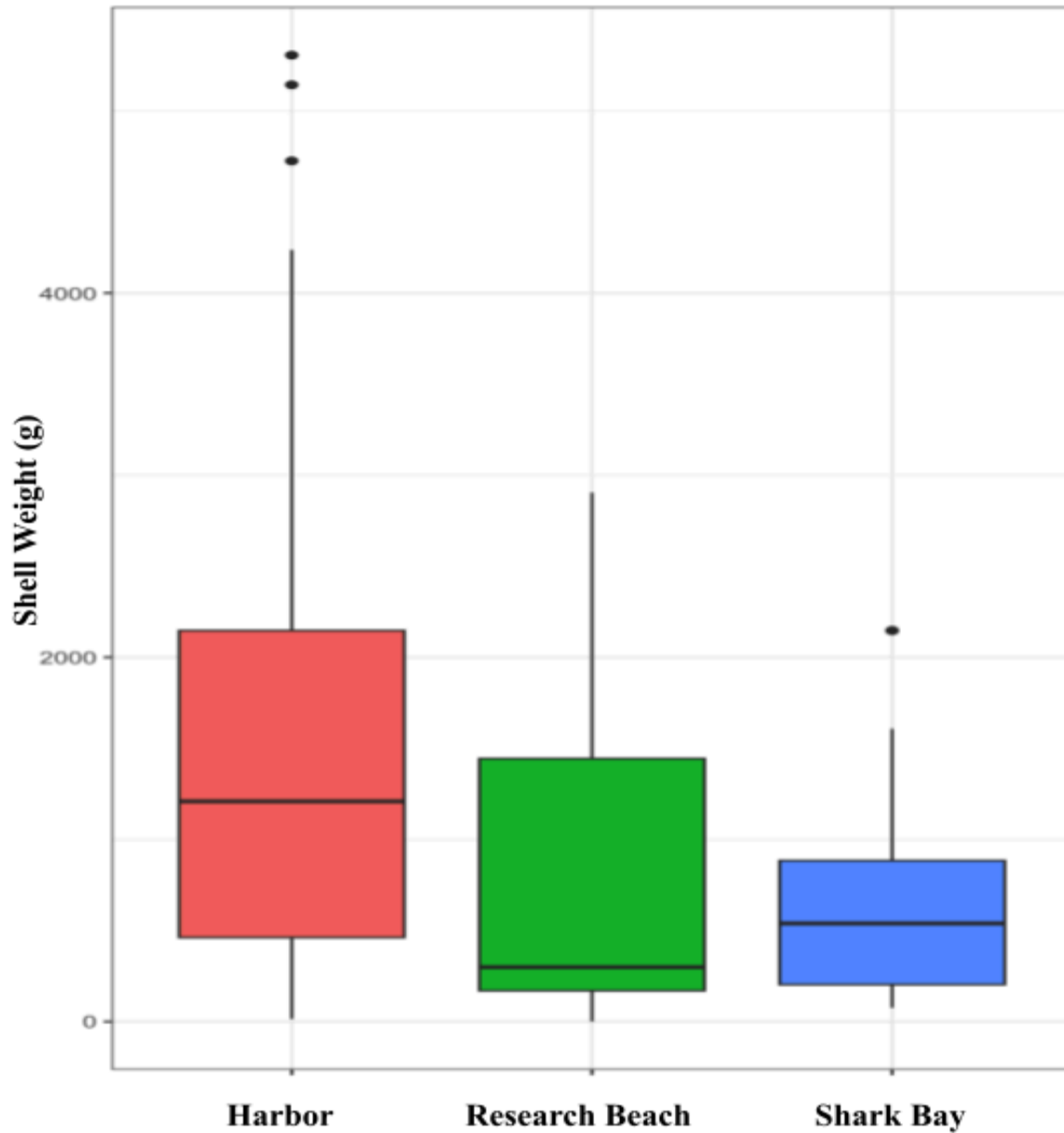


Fig. 5. Comparison of calculated shell weight (g) of *T. maxima* across the three sites: the Harbor (disturbed), Research Beach and Shark Bay (undisturbed).

Dry tissue biomass violated normality and variance like shell weight. Similarly, a significant difference between biomass was seen across the three sites (Kruskal-Wallis $\chi^2(2) = 14.27$, $p = 0.0008$). Approximated dry tissue biomass was significantly greater at the Harbor compared to Research Beach ($p = 0.0027$) and Shark Bay ($p = 0.0027$), as determined by a

Pairwise Wilcox test. Dry tissue biomass did not significantly differ between Research Beach and Shark Bay ($p = 0.49$) (Fig. 6).

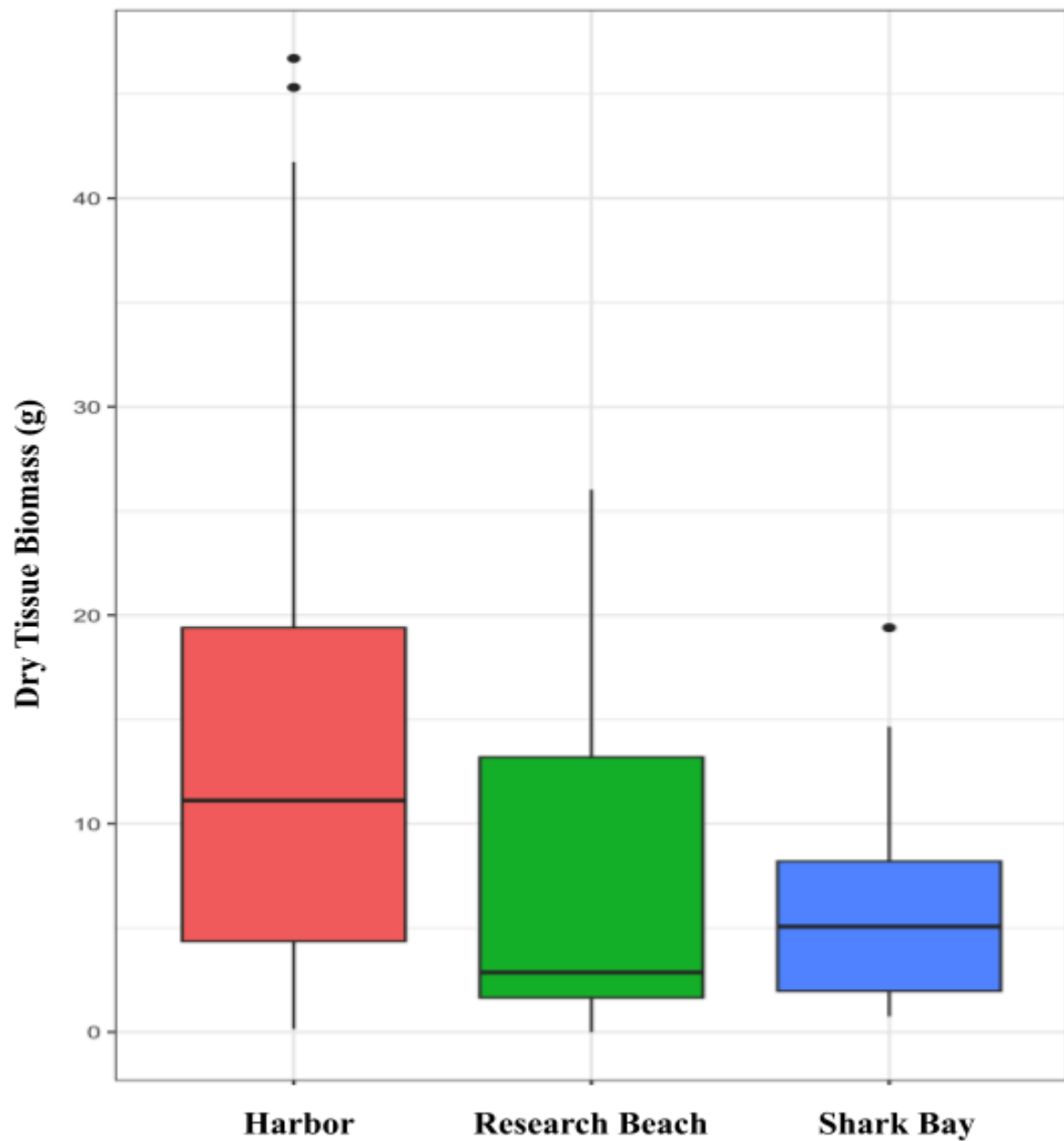


Fig. 6. Comparison of calculated dry tissue biomass (g) of *T. Maxima* across the three sites: the Harbor (disturbed), Research Beach and Shark Bay (undisturbed).

The ratio of juveniles to adults varied significantly across all 3 locations, as determined by a Pearson's chi-squared test ($\chi^2(2) = 12.56$, $p = 0.001874$) (Table 1). At the Harbor, 71.4% of individuals counted were adults, while 28.6% were juveniles. At Research Beach, 33.3% of the individuals counted were adults, leaving 66.7% as juveniles. Similarly, Shark Bay had 45.5% adults, and 54.5% juveniles.

Table 1. Recorded number (n) and respective percentage of *T. maxima* juveniles and adults across the three sites: the Harbor (disturbed), Research Beach and Shark Bay (undisturbed).

Site	Juvenile (n, %)	Adult (n, %)	Total n
Harbor	14 (28.6%)	35 (71.4%)	49
Research Beach	22 (66.7%)	11 (33.3%)	33
Shark bay	18 (54.5%)	15 (45.5%)	33

Discussion

This study compared *T. maxima* populations across a disturbed site (the Harbor) and two undisturbed sites (Shark Bay and Research Beach) on Heron Island, examining variation in density, size, biomass, and age composition. There was no significant difference between the density of *T. maxima* across all three sites. However, there was a notable difference between shell length, weight, dry tissue biomass, and the ratio of adults to juveniles. *T. maxima* were larger and heavier near the Harbor, with a greater proportion of adults compared to the two undisturbed sites (Shark Bay and Research Beach). There were no significant differences between these variables across Shark Bay and Research Beach.

No significant differences in density suggests that Harbor disturbance does not affect *T. maxima* survival. However, non-random sampling methods could explain this result. Transects were placed with a bias toward a larger clam density to ensure adequate sample size, which would reduce the representativeness of our dataset in reflecting the density of all three sites. Additionally, high winds and current shifted transect lines during sampling at Shark Bay, which could unintentionally increase the area surveyed than what was recorded, hindering projected density calculations.

Both Shark Bay and Research Beach lacked significant differences between all measured variables, which likely reflects their similarity in landscape and coral composition. Both sites were also categorized as undisturbed due to their distance from the Harbor. Therefore it was expected to see minimal differences between the two sites.

The significantly greater shell length, weight, and dry tissue biomass at the Harbor align with prior studies that suggested that increased water flow from harbor activity would increase nutrient enrichment (Rippe *et al.* 2024), and thus enhance *T. maxima* shell length growth (Ambariyanto 2007). Dredging of the harbor and boat activity likely increases dissolved nutrient availability, promoting calcification and tissue growth more than in undisturbed areas.

Although larger tissue biomass has a lower density of zooxanthellae per gram of tissue, and thus lower photosynthetic rates (Griffiths and Klumpp 1996), high water flow from the Harbor may improve endosymbiotic performance. An additional study suggests that under high flow speeds, coral endosymbionts can maintain photosynthetic activity for longer due to increased transfer of dissolved metabolites and nutrients (Page *et al.* 2021). This supports the maintenance of photosynthetic rate and growth, despite seemingly unfavorable conditions (Page *et al.* 2021). Given the physiological similarities between coral and giant clams (Uchida *et al.*

2025), increased water flow may also help sustain *T. maxima* growth. This may explain the larger growth along the Harbor despite the potential consequences suggested by increased turbidity (Hart *et al.* 1998).

The lower proportions of juveniles at the Harbor coincide with prior studies that hypothesize lower rates of juvenile recruitment in high flow environments due to high rates of sedimentation (Van Wynsberg *et al.* 2016; Rippe *et al.* 2024; Triandiza *et al.* 2019). Limited recruitment and juvenile settlement near the Harbor may lead to an aging population of *T. maxima* over time (Triandiza *et al.* 2019). The dominance of larger, aging *T. maxima* may increase local carbonate deposition (Mohammed *et al.* 2019) and nutrient recycling (Triandiza *et al.* 2019, Neo *et al.* 2015). When dead, their heavy shells contribute to the calcium carbonate deposition in their respective reef ecosystem by adding coarse carbonate sediment as their shell breaks down (Neo *et al.* 2015). Larger biomass at the Harbor can suggest larger nutrient recycling as the mantle tissue of the clams produces a ready source of inorganic and organic nutrients to reef waters (Chan *et al.* 2008) by releasing fecal pellets. Thus, larger biomass likely means larger fecal expulsion, which enhances nutrients for other endosymbiotic reef species and increases local reef productivity (Neo *et al.* 2015). However, more research is needed to confirm the suggested conclusion by measuring fecal expulsion rate across different age groups of *T. maxima* on Heron Island.

In contrast, larger juvenile abundance at Research Beach and Shark Bay likely boosts *T. maxima* reef contribution through increased bioerosion from recruitment and settlement (Neo *et al.* 2015). Additionally, a higher photosynthetic rate associated with a smaller mantle size and higher zooxanthellae density contributes to increased growth rates (Griffiths and Klumpp 1996). Rapid juvenile growth through filter feeding contributes to nutrient recycling and removal in

these less-disturbed sites (Ambariyanto 2007; AbouElmaaty *et al.* 2024), arguably more so than the Harbor due to higher numbers, helping maintain healthy reef productivity.

T. maxima serve as indicators of environmental stress (Harahap *et al.* 2018) due to their close relation of growth and health to coral reef conditions (Lee *et al.* 2024). Reducing recruitment near the Harbor may reflect local habitat degradation as *T. maxima* juveniles prefer settlement within healthy reef ecosystems (Rippe *et al.* 2024). However, further studies are needed to quantify the direct relationship between Heron Island reef health and the local giant clam ecology.

Understanding these dynamics between human disturbance and giant clam ecology is important given their ecological importance to reefs. Continued research and monitoring of giant clam populations, like *T. maxima*, can provide valuable insight into the effects of human activity on the organisms and reef ecosystems. Future research is needed to quantify calcification and filtration rates across life stages in each site (disturbed and undisturbed) and determine nutrient enrichment thresholds in order to better understand their implications for reef productivity on Heron Island.

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University of California EAP – AI Declaration

I have used AI for my HIRS report and I have detailed below how AI was used in the planning, preparation, editing and/or writing of this report.

Tool	Use	Prompt	Section	Date
Elicit	Source Scouting	Find sources pertaining to Giant Clam Ecology and Sedimentation in the Great Barrier Reef	References	October 14, 2025
Elicit	Source and Methods Scouting	Find sources that have various methodology for measuring Giant Clam Abundance and Size on intertidal reefs	Methodology	October 14, 2025
Chat GPT	Coding using RStudio (Density Analysis)	I have the average clam density along 3 sites, what type of statistical test should I run to test for statistical significance. Give me a step by step explanation on how to carry out each test in RStudio.	Results	October 24, 2025
Chat GPT	Coding using RStudio (Shell Length, Weight and Biomass Analysis)	I have 3 sites, and collected a different number of clams at each site with a listed shell length, dry	Results	October 24, 2025

		biomass and shell weight. What statistical test can I run? Give a step by step explanation on how to carry out each test in RStudio.		
Chat GPT	Map creation using coordinates and RStudio	I have a bunch of coordinates for places we surveyed and measured clams, how can I create a visual map in RStudio of their location on a satellite map.	Methodology	October 26, 2025
Chat GPT	References Formatting	Attached is the criteria for the Marine and Freshwater Journal References Guide. Format my sources to match the guide in alphabetical order.	References	November 3, 2025
Chat GPT	Editing and Revising	Point out where the report matches the rubric and areas to improve.	Whole paper	November 4, 2025